QC STORY PROJECT

PROJECT TITLE :- To eliminate fuel tank bolt not fit/loose concern from all vehicle models.

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	01/07/2019	01/07/2019
Completion Date	10/12/2019	20/11/2019

Current Status:

CLOSED

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

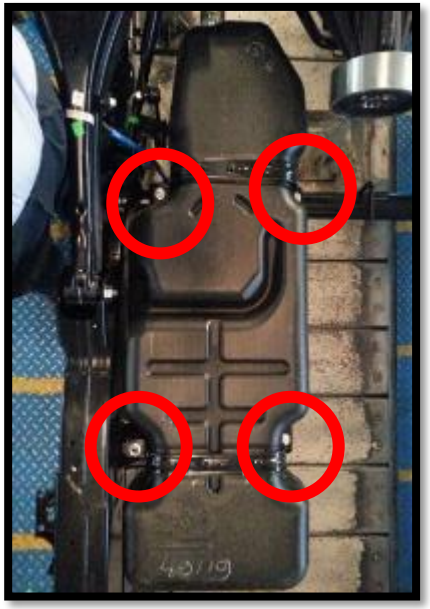
Cross Functional Team

Sr no.	Description	Name	Designation	Dept	Phone	Email
1	Team Leader	Mandar Padwal	Asst. Manager	MFG	8999392642	Padwal.mandar@ Mahindra.com
2	Mentor	Taj Shaikh	Sr. Manager	STCF	8378991344	Shaikh.taj@Mahindra.com
3	Member 1	Ajit Chaudhari	Dy. Manager	MFG	7588616388	Chaudhari.ajit@Mahindra.com
4	Member 2	Santosh Jadhav	Dy. Manager	MQA	9423309400	Jadhav.santosh8@ MAHINDRA.COM
5	Member 4	Shreeyash Nage	Asst. Manager	MQA	9167382673	Nage.shreeyash@ mahindra.com
6	Member 3	Darshak Thakkar	Dy. Mgr	SQA	7720074994	Thakkar.darshak@ mahindra.com

1. Select the Theme

Brief description of Problem

- What : Fuel tank bolt loose.
- Where : At the Chassis Line.
- When : Since APR-19
- Who : By stage operator, Buy off.
- Why : Bolts are not tightened on tank.
- How : Fuel Tank Bolt Loose- Freq 95 in June 19.

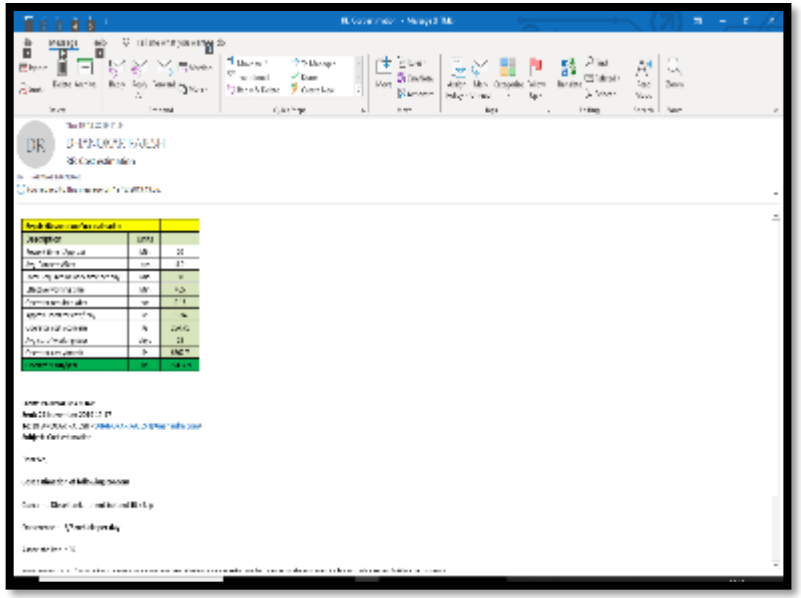


2. Justify the Choice (Reasons)

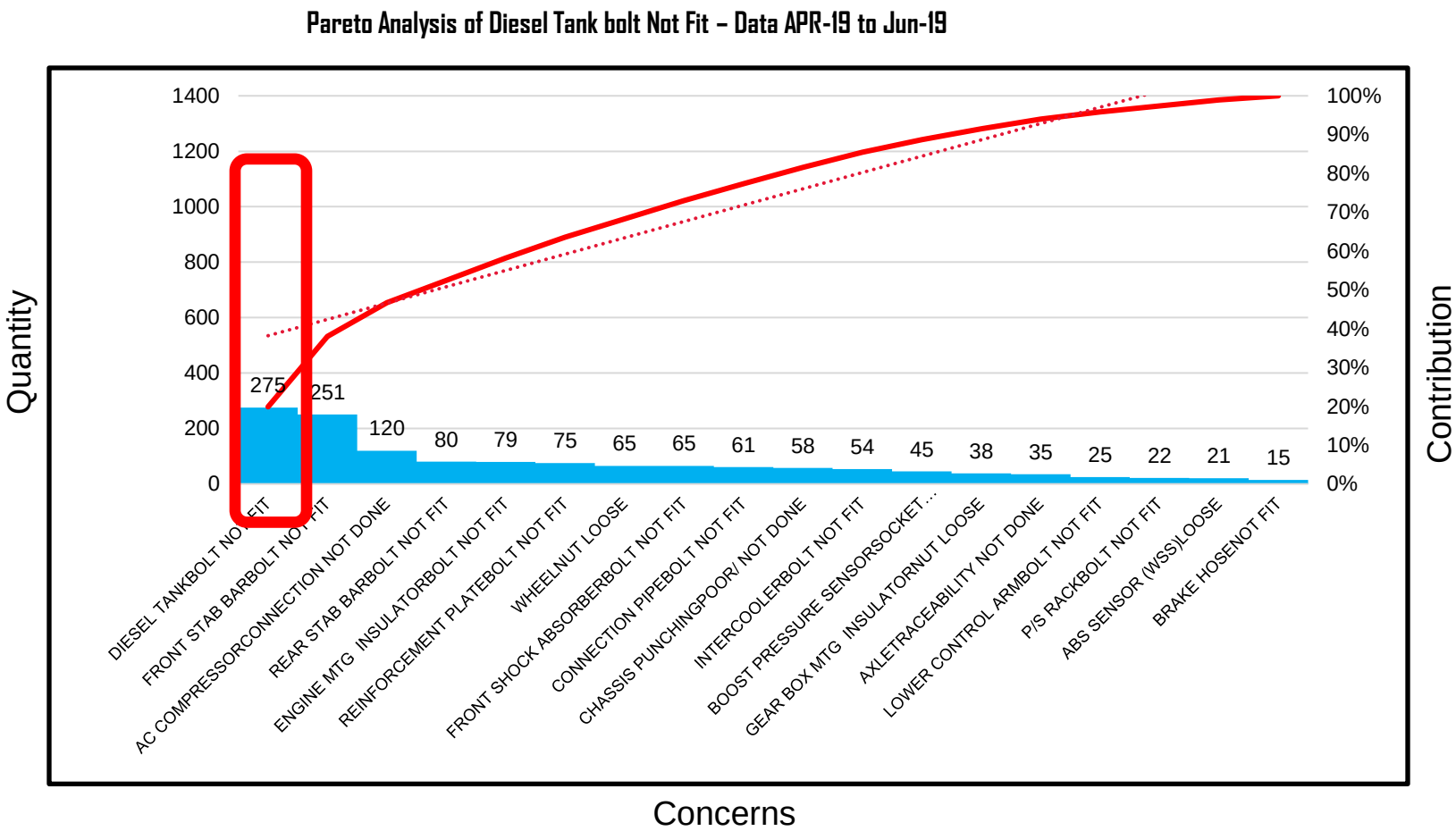
Avg. 92 Cases reported per month, for rework

Rework time 20 mins for each vehicle by 2 operator.

Description	Units	Value
Rework Time Approx.	Min.	20
Avg. Concern per day	Nos.	3.0
Total Time required per day	Mins.	60
Effective working Time	Mins.	465
Operator Required per day	Nos.	0.13
Approx. Operator Cost/day	Rs.	1974
Operator Cost/Concern	Rs.	254.71
Avg. No of Working days	Days	25
Operator Cost /Month	Rs.	6367.7
Operator Cost /year	Rs.	76412.9

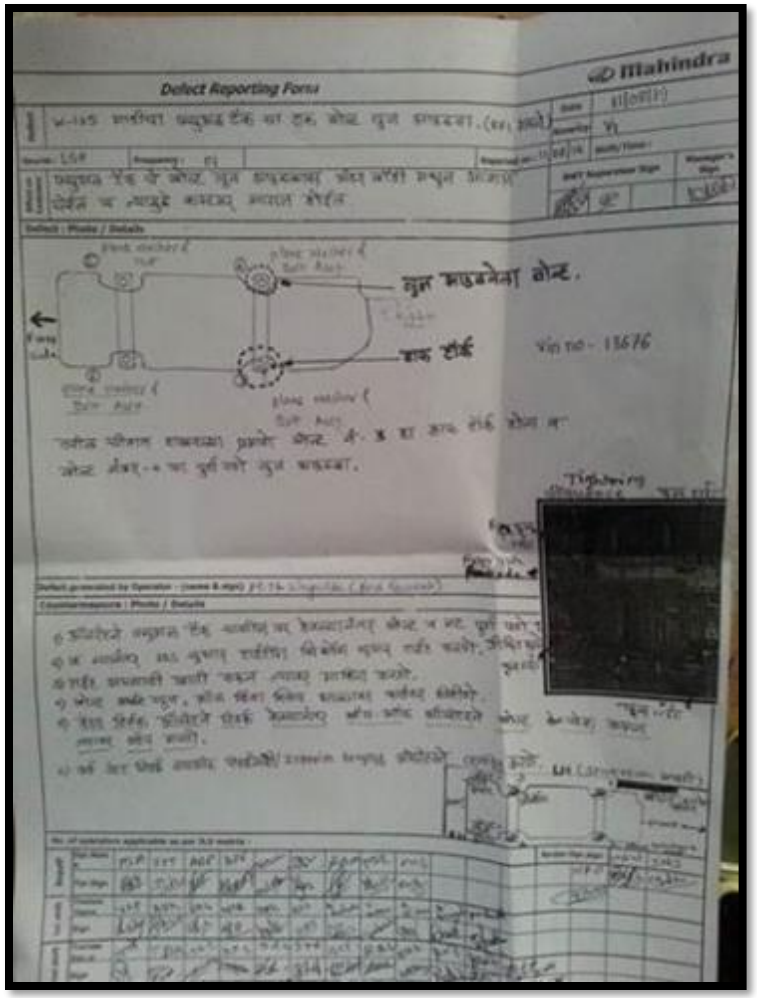


2. Justify the Choice (Reasons)



Fuel Tank Bolt is numbering in 1st Position in PARETO.

2.a. Immediate Containment Action



- DRF given to operators for Concern detection at Fuel Tank & Buy off at Chassis

3. Understand the Current Situation

Observation of Process

Chassis In → Fuel tank Fitment → Engine Drop → Buyoff

Concern Generation Stage

Concern Rectification Stage

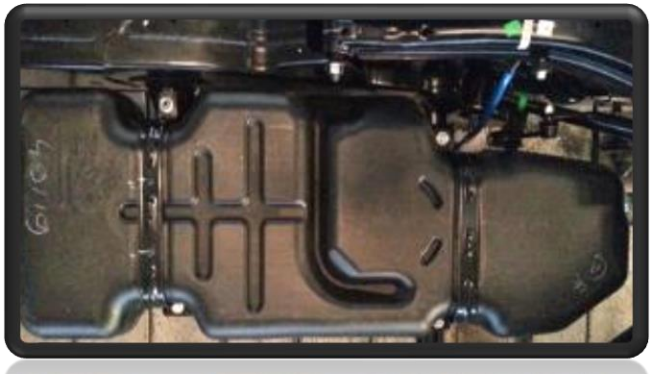
↓

FAI → CAI →

3. Understand the Current Situation

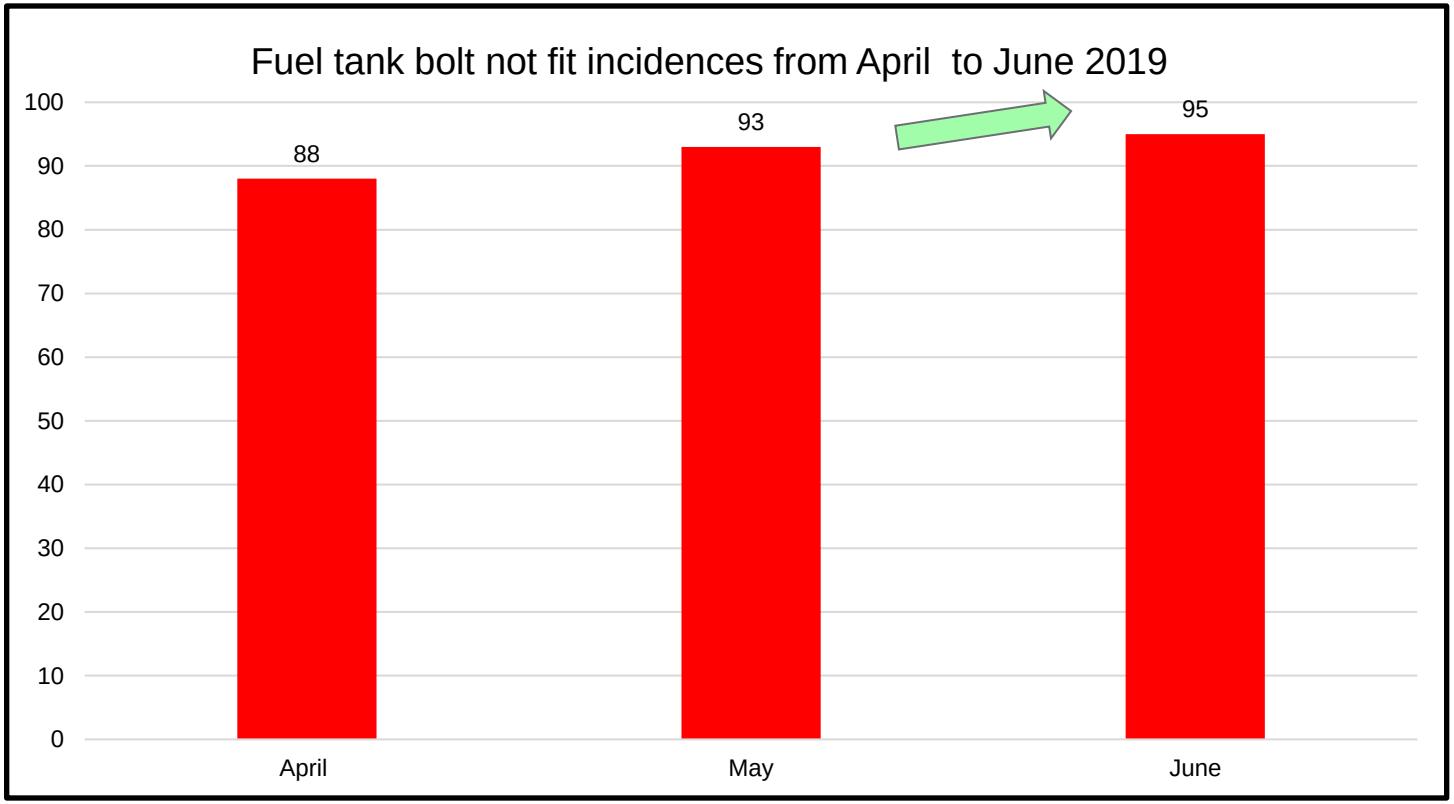
Observation of Process

4 Mounting of M8 nuts on the Chassis



Holes provided on the Fuel Tank Strap used for Fastening

3. Understand the Current Situation



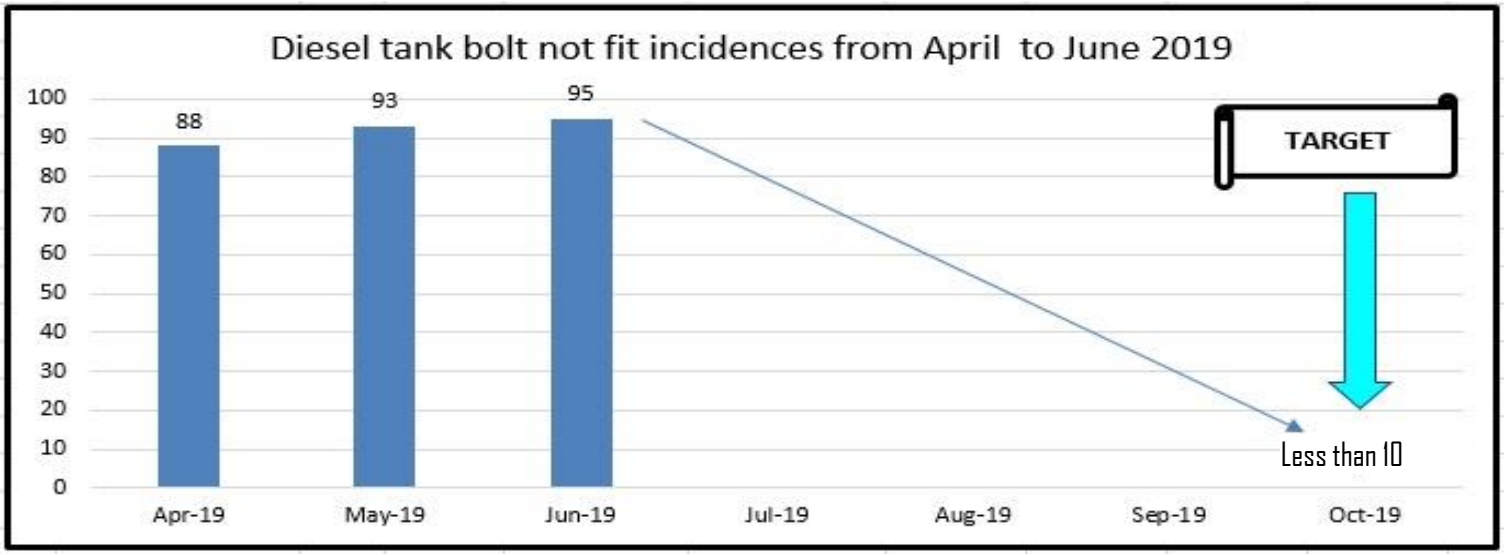
3. Understand the Current Situation

L0 Analysis for Fuel tank fitment Process

4. Select the Target

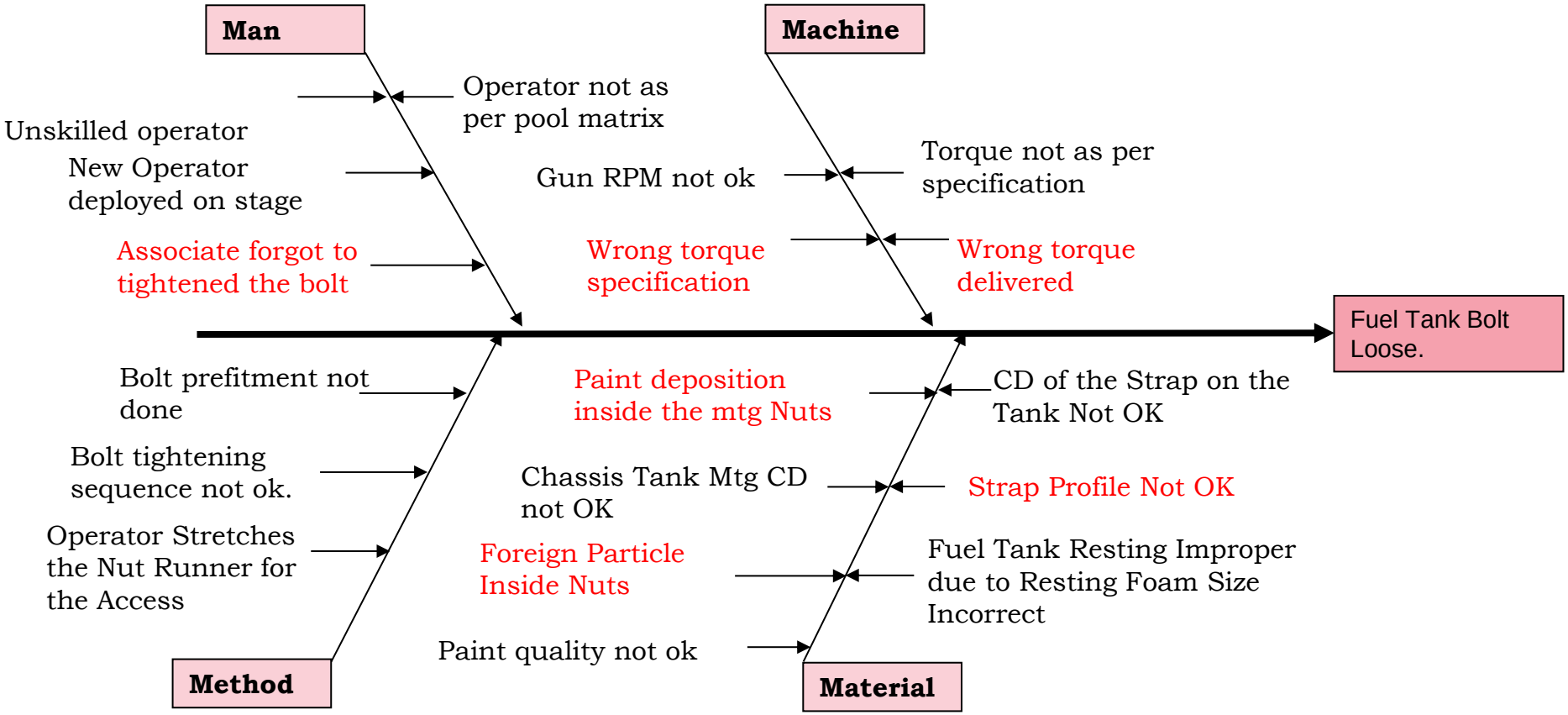
Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level :	95 Cases reported at Buyoff based on data of June '19
Target :	To eliminate Fuel tank Bolt loose concern by Oct '19



5. Analysis

5.1 Identify possible causes: Fish bone diagram.



18 possible causes and 6 probable causes identified.

5. Analysis

5.1 List of Possible causes

S N	Probable Cause	Taj	Ajit	Santosh	Shreeyash	Darshak	Mandar	Total
1	Unskilled operator	0	0	1	1	0	1	3
2	New Operator deployed on stage	1	1	1	1	1	3	8
3	Associate forgot to tightened the bolt	9	1	9	9	1	9	38
4	Operator not as per pool matrix	0	0	3	0	3	3	9
5	Gun RPM not ok	0	0	0	1	0	1	2
6	Torque not as per specification	3	1	1	1	0	3	9
7	Wrong torque specification	9	1	9	1	3	9	32
8	Operator stretches nut runner for the access	0	0	3	1	0	3	7
9	Bolt pre fitment not done	0	1	1	3	0	3	8
10	Bolt tightening sequence not ok	1	1	0	1	0	1	4
11	Paint deposition inside the mtg Nuts	0	9	9	9	3	9	39
12	Chassis Tank Mtg CD not OK	0	0	0	0	0	3	3
13	Foreign Particle inside Nuts	1	3	9	3	9	9	35
14	Paint quality not ok	3	3	1	1	0	3	14
15	Wrong torque delivered	1	3	3	9	1	9	26
16	Strap Profile Not OK	0	9	9	3	3	3	27
17	Fuel Tank Resting Improper due to Resting Foam Size Incorrect	0	0	1	3	0	3	7
18	CD of the strap on the tank not ok	0	0	0	0	1	3	4

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 27.

5.1 Test of Hypothesis

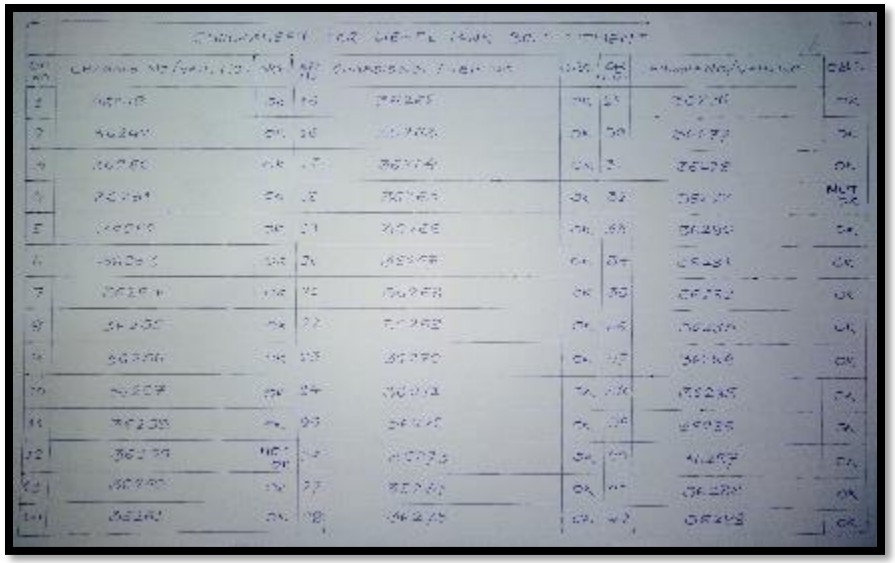
Confirm the Probable cause:

Sr.No	Probable Cause
1	Associate forgot to tightened the bolt
2	Wrong torque specification
3	Paint deposition inside the mounting Nuts
4	Foreign Particle inside the Nuts
5	Wrong torque delivered
6	Strap Profile Not OK

5.1 Test of Hypothesis

5.2 Confirm the Probable cause:

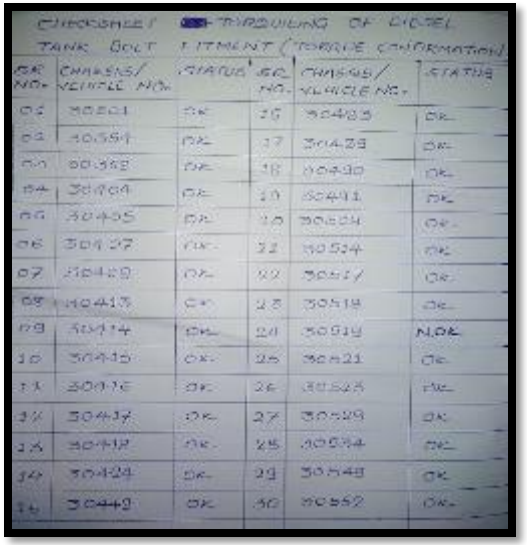
S.N.	Probable Cause	Testing method	Observations	Conclusion
1	Associate forgot to tightened the bolt	Set of 42 chassis check visually and also with brake back torque wrench of 30Nm, and check whether associate forgot to tightened the bolt or not.	2 Out of 42 chassis found with bolt not tightened/ under torque.	**Valid**



5.1 Test of Hypothesis

5.2 Confirm the Probable cause:

S.N.	Probable Cause	Testing method	Observations	Conclusion
2	Wrong torque delivered	Set of 30 chassis check using brake back torque wrench of 30Nm, and check whether correct tool used by associate or not.	1 out of 30 vehicles tank bolt slipped due to over torqueing on tank bolts	**Valid**

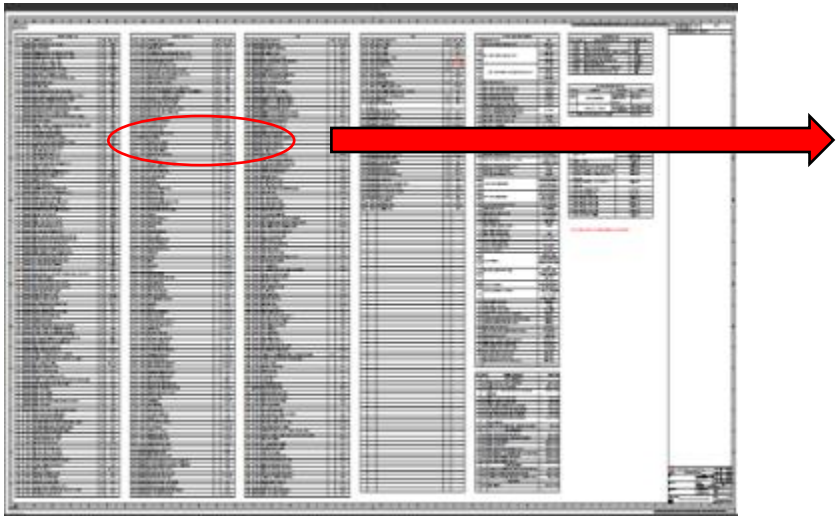


SR NO.	CHASSIS/ VEHICLE NO.	STATUS	SR. NO.	CHASSIS/ VEHICLE NO.	STATUS
01	30501	OK	16	30483	OK
02	30504	OK	17	30428	OK
03	30502	OK	18	30490	OK
04	30404	OK	19	30491	OK
05	30405	OK	20	30504	OK
06	30427	OK	21	30534	OK
07	30420	OK	22	30517	OK
08	30413	OK	23	30518	OK
09	30414	OK	24	30519	NOK
10	30415	OK	25	30521	OK
11	30416	OK	26	30525	OK
12	30417	OK	27	30529	OK
13	30418	OK	28	30534	OK
14	30424	OK	29	30548	OK
15	30442	OK	30	30552	OK

5.1 Test of Hypothesis

5.2 Confirm the Probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
3	Wrong torque spect.	Torque specifications checked with torque engineering specifications. (30±5Nm for Scorpio & 50±5Nm for Marrazzo)	Torque specifications checked with torque engineering specifications found as per drawing. Actual :- Ranging from 29.53 – 30.11 Nm (Scorpio) And 53.95-55.96 Nm (Marrazzo) As per ES :- 30±5 Nm and 50±5 Nm	**Invalid**



124	Chassis	Stabilizer Mtg	45±8
125	Chassis	LCA Joint	230±20
126	Chassis	UCA Joint	90±10
127	Chassis	Multilink	150±15
128	Chassis	Fuel Tank	30±5

5.1 Test of Hypothesis

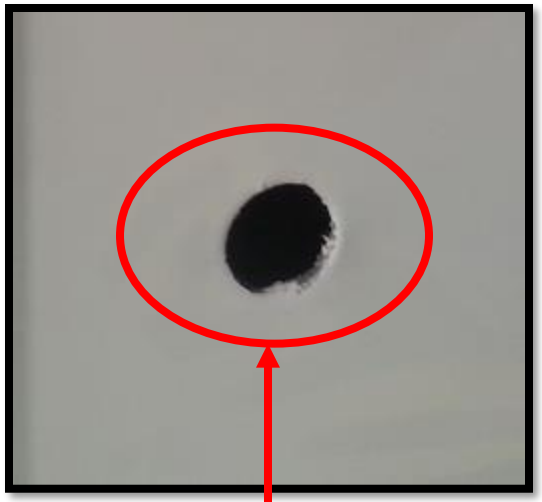
5.2 Confirm the Probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
4	Paint deposition inside the mounting Nuts	Sample Test of 15 Chassis for pre fitment test	Out of 15 Chassis checked 5 were found for hard for pre fitment	**Valid**

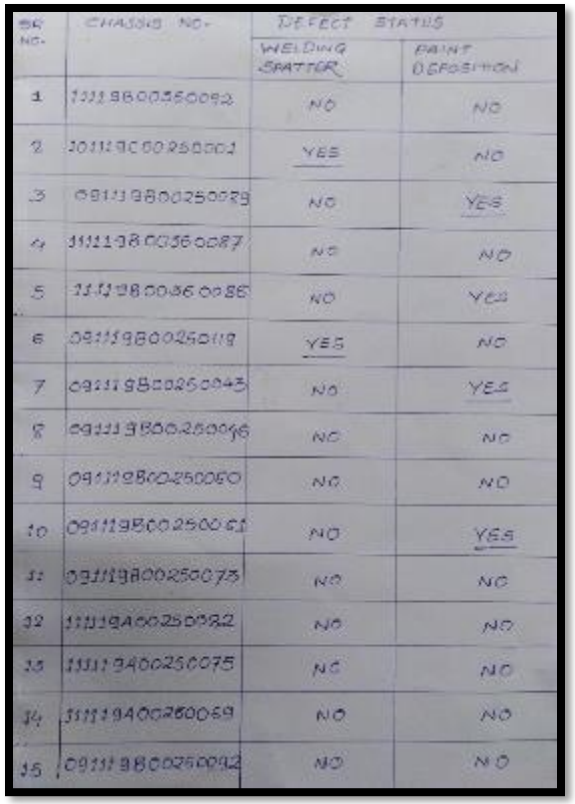
Chassis physically check for paint deposition concern

5.1 Test of Hypothesis

5.2 Confirm the Probable cause:



Foreign Particle Weld Particle Observed



SR NO.	CHASSIS NO.	DEFECT STATUS	
		WELDING SPATTER	PAINT DEPOSITION
1	1111980036 0092	NO	NO
2	1011190 00250001	YES	NO
3	09111980025 0029	NO	YES
4	1111198 0036 0087	NO	NO
5	11 11198 0036 0088	NO	YES
6	09111980025 0119	YES	NO
7	09111 98 00250043	NO	YES
8	09111 9800250096	NO	NO
9	09111980025 0060	NO	NO
10	09111980025 00 61	NO	YES
11	09111980025 0075	NO	NO
12	1111198 0025 0082	NO	NO
13	1111 1940025 0075	NO	NO
14	1111 1940025 0069	NO	NO
15	09111 980025 0092	NO	NO

5.1 Test of Hypothesis

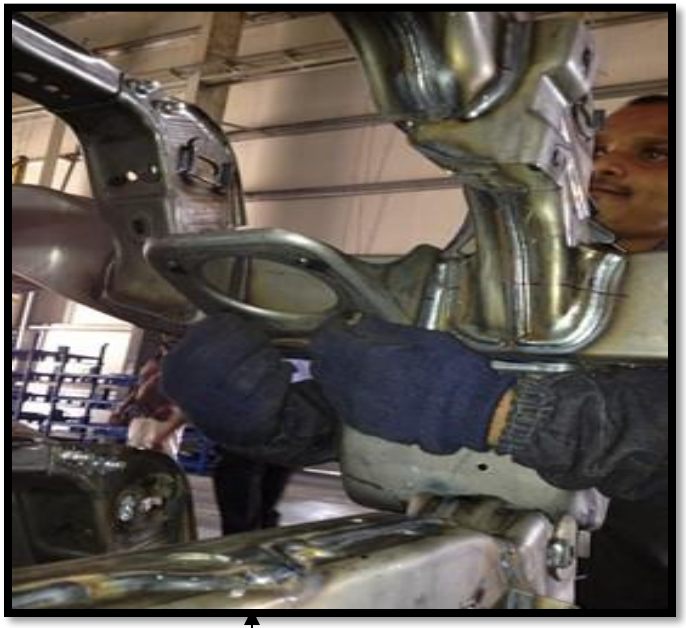
5.2 Confirm the Probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
5	Foreign Particle inside the Nuts	Sample Test of 15 Chassis for pre fitment test	Out of 15 Chassis checked 2 were found with Weld spatter	Valid

Detail observation report in next slide…

5.1 Test of Hypothesis

5.2 Confirm the Probable cause:



Chassis frame threads quality checking at supplier end with bolt pre fitment.

5.1 Test of Hypothesis

5.2 Confirm the Probable cause:

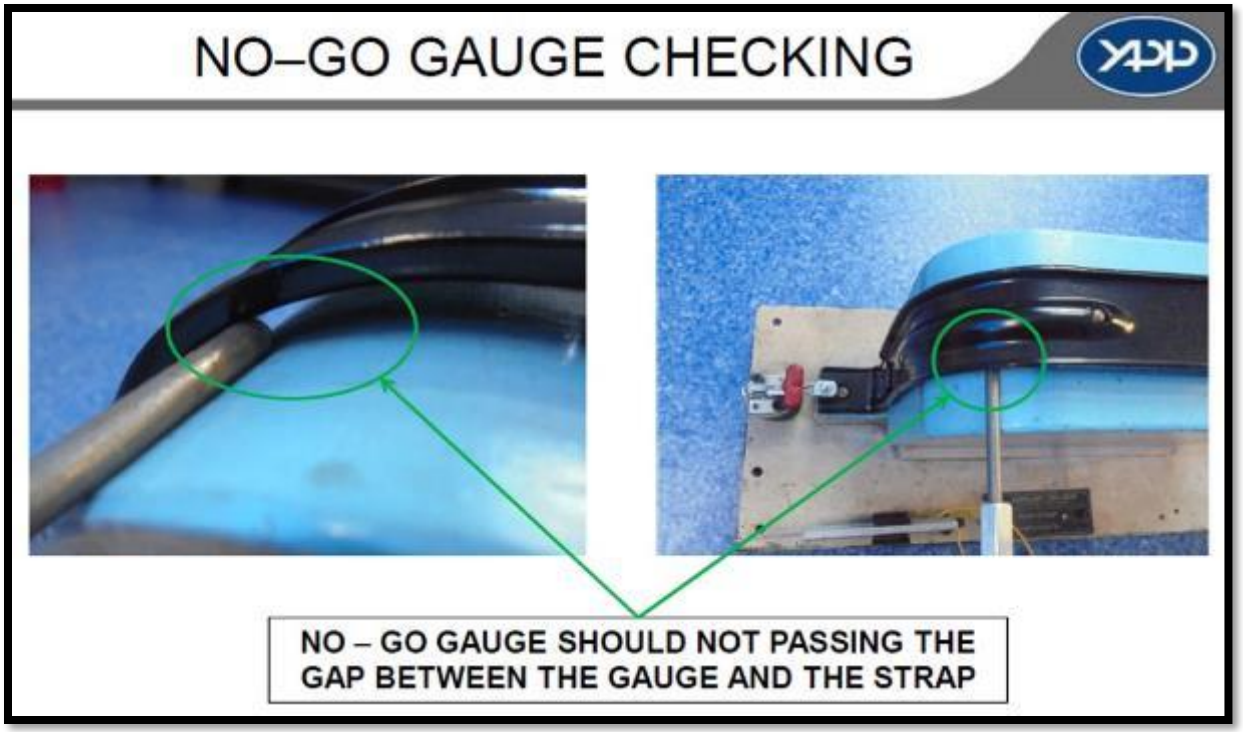
No	Probable Cause	Testing method	Observations	Conclusion
6	Strap Profile Not OK	Sample Test of 15 strap for profile test	Out of 15 strap checked all found OK	**Invalid**

- MOUNT THE STRAP ON THE SHAPE GAUGE
- CHECK WHETHER ALL HOLES ARE MATCHING ON THE SHAPE GAUGE PINS
- IF HOLES ARE MACHING WITH PINS THEN, CLAMP THE END TOGGLE PINS ON THE STRAP
- CHECK WHETHER GO & NO GO GAGUE ARE ANSWERING OR NOT
- USE THE SAME PROCEDURE FOR BOTH THE STRAPS

Further observations on next slide…

5.1 Test of Hypothesis

5.2 Confirm the Probable cause:



WHY - WHY Analysis

5.3 Identify the root causes : (Use Why-Why)

Sr. No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION
1	**Paint deposition inside the mtg. Nuts**	Bolt Pre Fitment difficult	Bolt Pre fitment Obstruction due foreign Particle	Bolt Pre fitment threading surface irregular	Threading Surface Paint deposition Excess	Nut Threading to be protected Paint Deposition by Masking
2	**Foreign Particle inside the Nuts**	Bolt Pre Fitment difficult	Presence of foreign Particle	Bolt Pre fitment threading surface irregular	Weld Spatter sticks on the Threading Surface	Nut Threading to be protected from Weld Spatter by Masking
3	**Wrong torque delivered**	Skip from associate as there are two tools used	Two similar looking guns involved in torqueing	Two diff. torque values delivered from two diff. tools as per vehicle models	Both the tools are pneumatic angular type hence two diff. torque can not be possible to achieve on single tool	Tool selection mechanism required for different vehicle models
4	**Associate forgot to tightened the bolt**	Skip from associate	Associate did not notice whether bolt was tightened or not	There is not alert mechanism which indicates that bolt are tightened		Human error prevention mechanism to be introduce

6. Implement the corrective measures

Possible remedies Identified

Sr. No	Valid Probable Cause	Remedy Identified	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Threading Surface Paint deposition Excess	Nut Threading to be protected Paint Deposition by Masking	Medium	Medium	No	Low	Acceptable
2	Weld Spatter sticks on the Threading Surface	Nut Threading to be protected from Weld Spatter by Masking	Medium	Medium	No	Low	Acceptable
3	Both the tools are pneumatic angular type hence two diff. torque can not be possible to achieve on single tool	Tool selection mechanism required for different vehicle models	High	Medium	No	Low	Acceptable
4	There is not alert mechanism which indicates that bolt are tightened	Human error prevention mechanism to be introduce	High	Medium	No	Low	Acceptable

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Paint deposition inside the mtg. Nuts	**Nut Threading to be protected Paint Deposition by Masking**	25 Aug 2019	Shailesh	Closed

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Foreign particle inside nut	**Nut Threading to be protected from weld spatter by Masking**	20 Aug 2019	Shailesh	Closed

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Wrong torque delivered	**Tool selection Poka-yoke to be initiate**	25 Sep 2019	Sayali More	Closed

Poka-yoke scanner

Model-wise diesel tank barcode scanning

Automatic tool selection as per vehicle model, Only required tool operates out of two tools

Correct torque delivered on fuel tank bolt

6. Implement the corrective measures

6.1 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Associate forgot to tightened the bolt	**Poka-yoke provided for toque confirmation**	20 Sep 2019	Sayali More	Closed

Yellow Indicator flashing on tower

Fuel tank barcode scanning

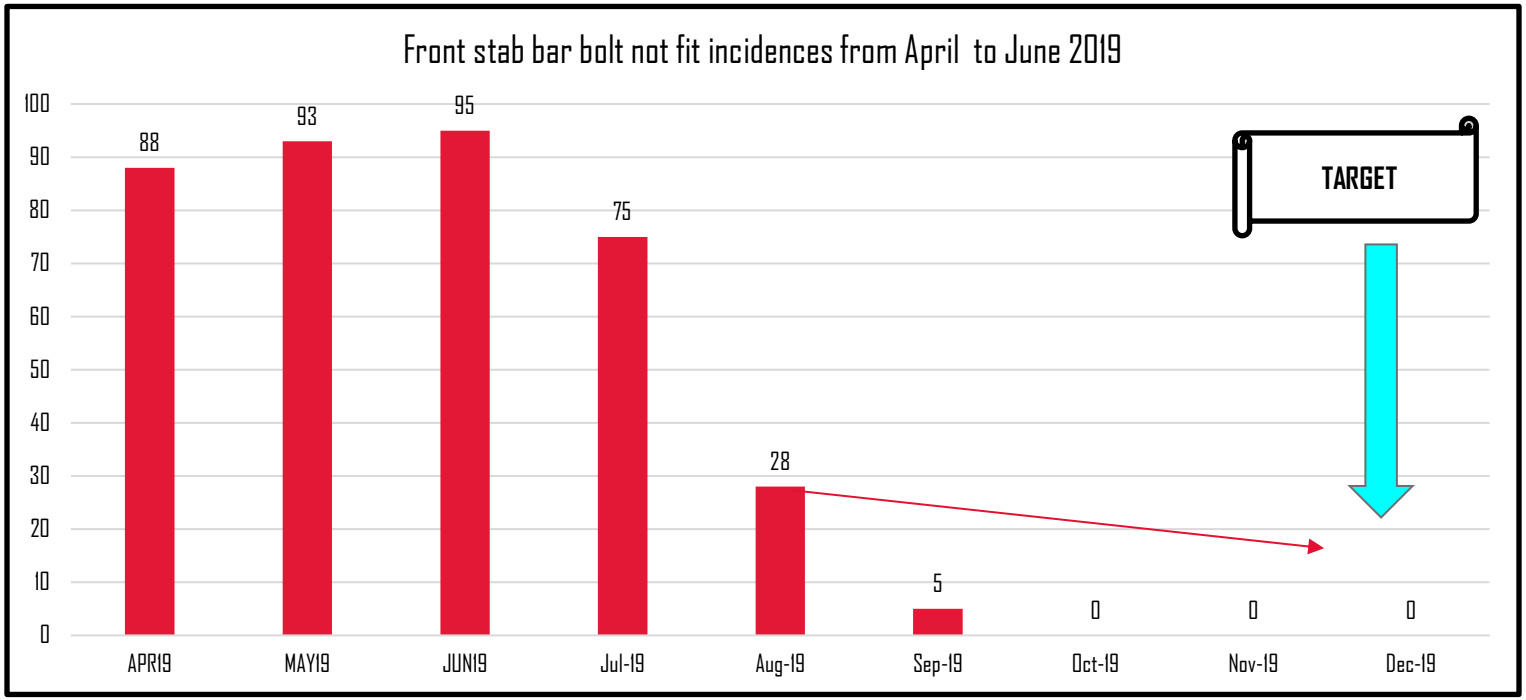
Intermittent Green indicator flashing

Required tool got selected and Tank bolt tightening

Steady green indication means all torque done (Confirmed)

7. Check and Confirm the results & effects

7.1 Verify Results



7. Check and Confirm the results & effects

7.2 Full Implementation Plan:

Action Plan	Target Date	Resp.	Status
Nut Threading to be protected Paint Deposition by Masking	25th Aug 2019	Shailesh Kumar	Closed
Nut Threading to be protected from Weld Spatter by masking	20th Aug 2019	Shailesh Kumar	Closed
Gun selection poka-yoke initiated	25th Sep 2019	Sayali More	Closed
Poka-yoke provided for human error prevention	20th Sep 2019	Sayali More	Closed

7. Check and Confirm the results & effects

7.3 Tangible Benefits:

- Concern elimination from **95** to 0 cases. (i.e. average 3 per day)
- Rework man hour saving / Annual cost saving – 6367.7 / month – **76412.9 / year**

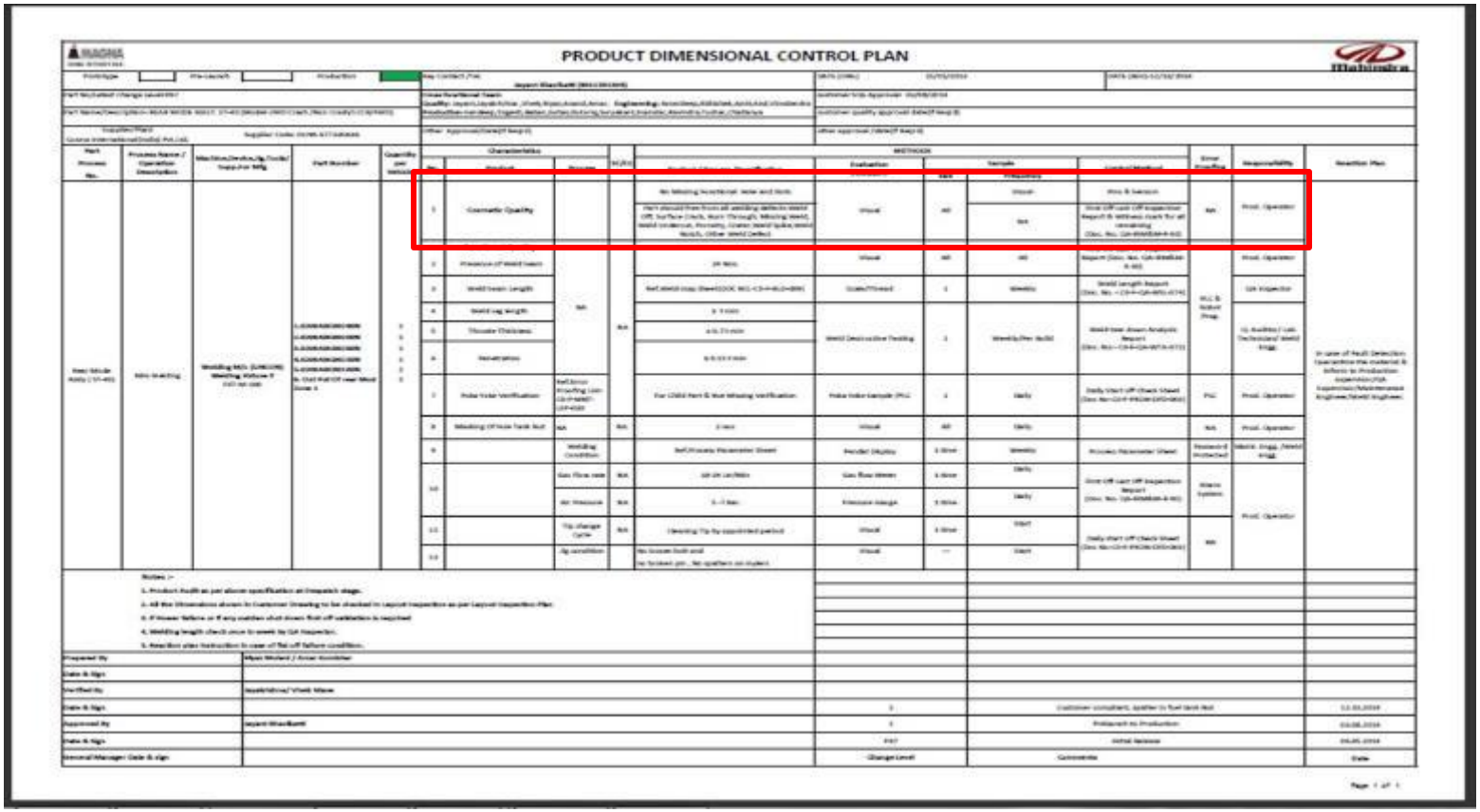
Description	Units	Value
Rework Time Approx.	Min.	20
Avg. Concern per day	Nos.	3.0
Total Time required per day	Mins.	60
Effective working Time	Mins.	465
Operator Required per day	Nos.	0.13
Approx. Operator Cost/day	Rs.	1974
Operator Cost/Concern	Rs.	254.71
Avg. No of Working days	Days	25
Operator Cost /Month	Rs.	6367.7
Operator Cost /year	Rs.	76412.9

- Customer satisfaction at stage Buyoff
- No FRC or FBO loss.
- Able to maintain built in quality due to zero rework.

8. Standardisation

8.1 Documents update:

1. Control Plan Magna Chassis Manufacturer:



PRODUCT DIMENSIONAL CONTROL PLAN

8. Standardisation

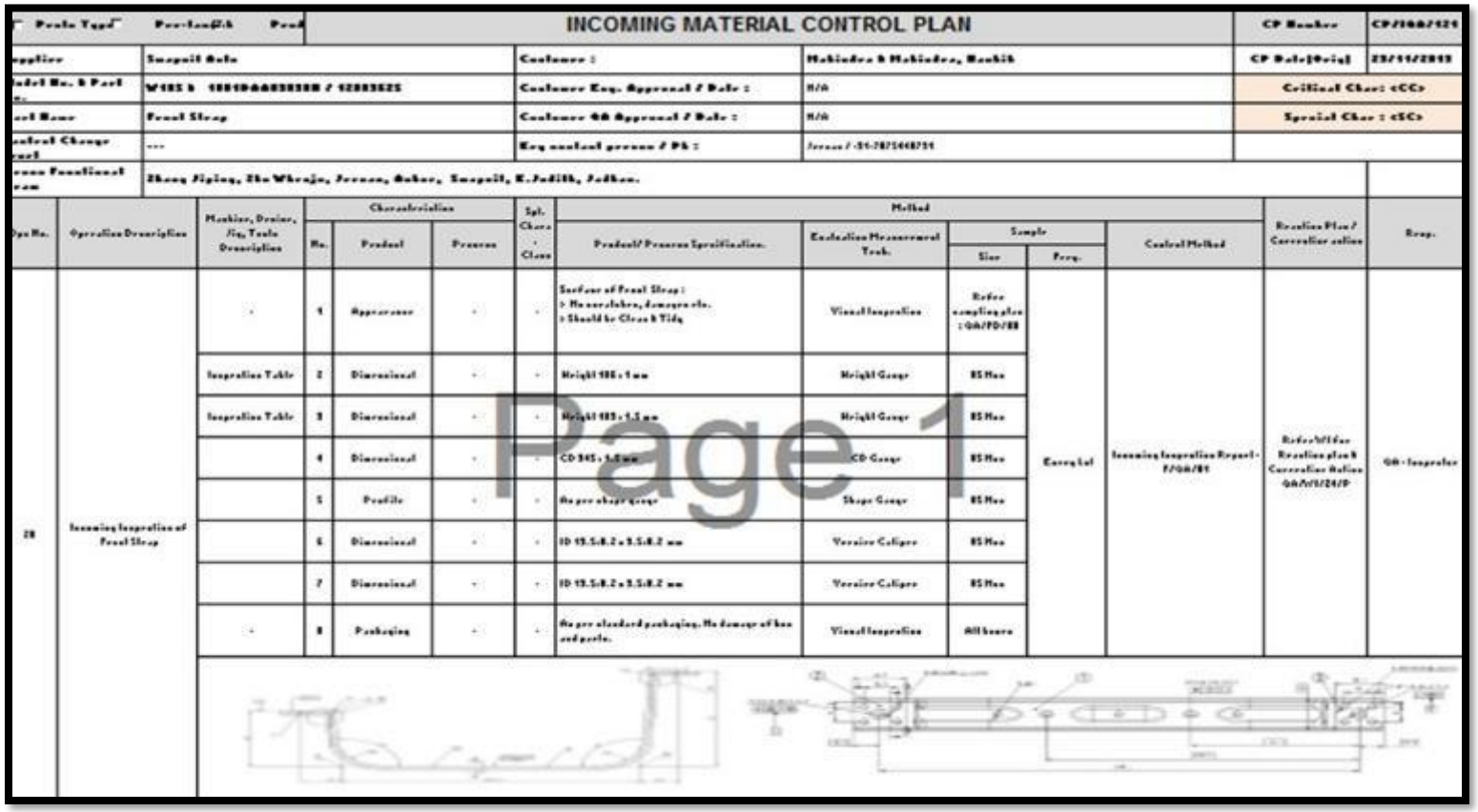
8.1 Documents update:

Part Number	per Vehicle	No.	Product	Process	SC/CC	Product / Process /Specification	Evaluation /Measure	Sample		Control Method	Error Proofing	Responsibility
								Size	Frequency			
						No Missing Functional Hole and Slots			Visual	Pins & Sensors		
		1	Cosmetic Quality			Part should free from all welding defects-Weld Off, Surface Crack, Burn Through, Missing Weld, Weld Undercut, Porosity, Crater,Weld Spike,Weld Notch, Other Weld Defect	Visual	All	NA	First Off Last Off Inspection Report & Witness mark for all remaining (Doc. No. QA-IRM&M-R 40)	NA	Prod. Operator
		2	Functional Quality Presence of Weld Seam	NA	NA	24 Nos.	Visual	All	All	First Off Last Off Inspection Report (Doc. No. QA-IRM&M-R 40)	PLC & Robot Prog.	Prod. Operator
		3	Weld Seam Length			Ref.Weld map Sheet(DOC NO.-CII-F-BLD-009)	Scale/Thread	1	Weekly	Weld Length Report (Doc. No. -:CII-F-QA-WSL-074)		QA Inspector
		4	Weld Leg length			≥ t min	Weld Destructive Testing	1	Weekly/Per Build	Weld tear down Analysis Report (Doc. No:- CII-F-QA-WTA-072)		Q. Auditor/ Lab Technician/ Weld Engg.
.0208ABC00290N	1	5	Throate Thickness			≥ 0.7 t min						
.0208ABC00200N	1											
.0208ABC00260N	1	6	Penetration			≥ 0.15 t min						
.0208ABC00230N	1											
.0208ABC00180N	2											
. Out Put Of rear Mod one 1	1	7	Poka Yoke Verification	Ref.Error Proofing List-CII-F-MNT-LEP-020		For Child Part & Nut Missing Verification	Poka Yoke Sample /PLC	1	Daily	Daily Start off Check Sheet (Doc No-CII-F-PRDN-DFD-003)	PLC	Prod. Operator
		8	Masking Of Fule Tank Nut	NA	NA	2 nos	Visual	All	Daily		NA	Prod. Operator
		9		Welding Condition		Ref.Process Parameter Sheet	Pendat Display	1 time	Weekly	Process Parameter Sheet	Password Protected	Maint. Engg. /Weld Engg.
		10		Gas Flow rate	NA	18-24 Ltr/Min	Gas flow Meter	1 time	Daily	First Off Last Off Inspection Report (Doc. No. QA-IRM&M-R 40)	Alarm System	Prod. Operator
				Air Pressure	NA	5 -7 Bar.	Pressure Gauge	1 time	Daily			
		11		Tip change Cycle	NA	Cleaning Tip by appointed period	Visual	1 time	Start	Daily start off Check Sheet	NA	

8. Standardisation

8.1 Documents update:

2. Control Plan YAPP India Fuel tank Manufacturer:



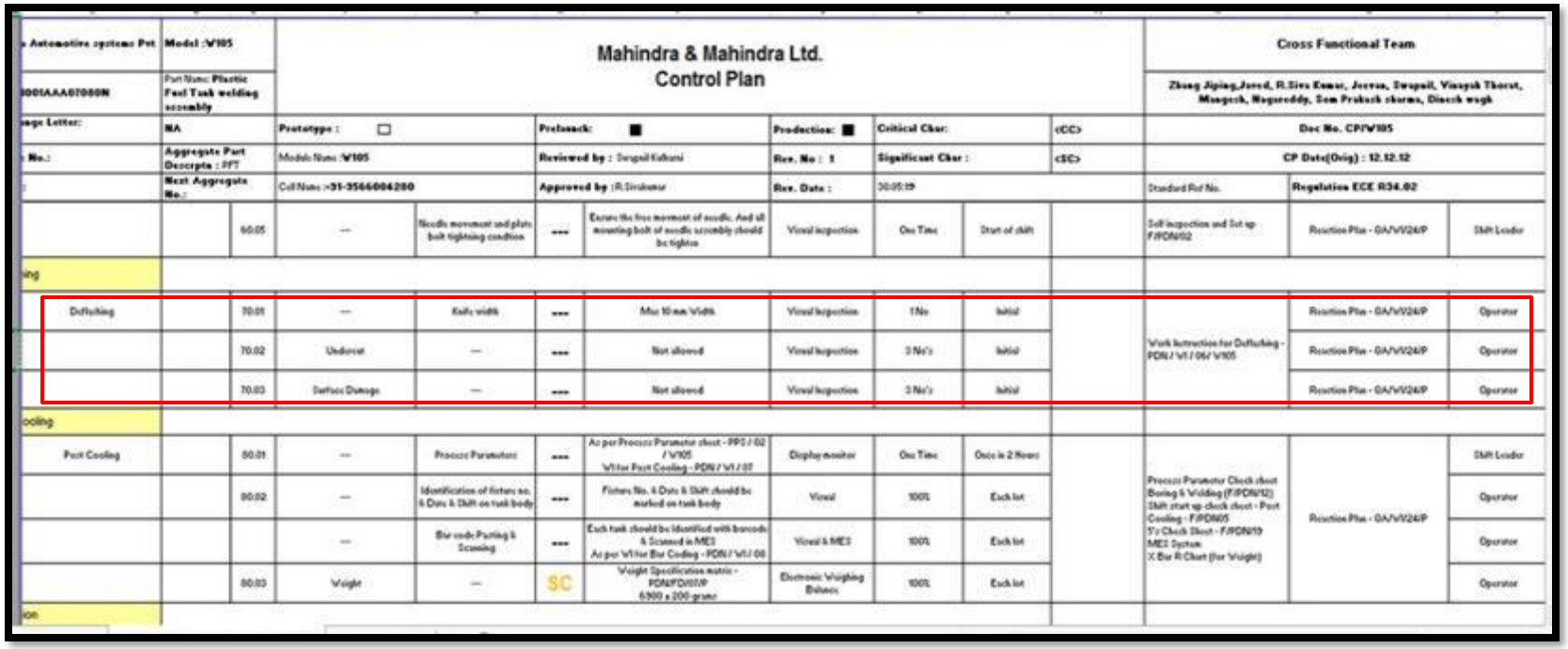
INCOMING MATERIAL CONTROL PLAN

Page 1

8. Standardisation

8.1 Documents update:

Action Plan	Target Dt	Resp.	Status
Control Plan Modified for Flash cutting at the Fuel Tank strap intersection	30 Aug 2019	Om Mali	Closed



Mahindra & Mahindra Ltd.
Control Plan

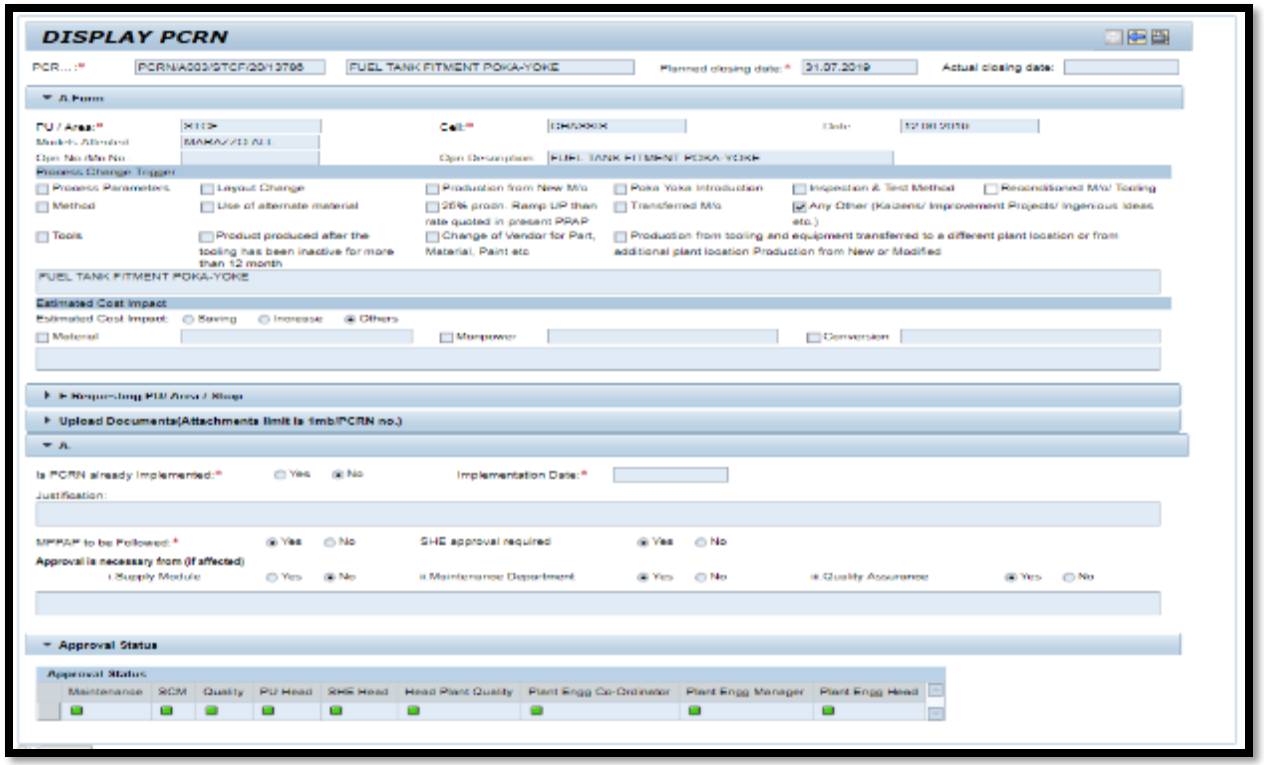
Cross Functional Team

Automotive systems Pvt	Model :W105												
001AAA07000N	Part Name: Plastic Fuel Tank welding assembly												
ange Letter:	NA		Prototype : ☐		Prelaunch: ■		Production: ■	Critical Char:		<CC>	Doc No. CP/W105		
No.:	Aggregate Part Descripte : PFT		Model Name :W105		Reviewed by : Swapnil Kulkarni		Rev. No : 1	Significant Char :		<SC>	CP Date(Orig) : 12.12.12		
	Next Aggregate No.:		Cell Name :-31-3566004200		Approved by : R.Sivakumar		Rev. Date :	30.05.19			Standard Ref No.	Regulation ECE R34.02	
		60.05	—	Needle movement and plate bolt tightening condition	---	Extract the free movement of needle. And all mounting bolt of needle assembly should be tighten	Visual inspection	One Time	Start of shift		Self inspection and Set up F/PDN/02	Reaction Plan - QA/V/24/P	Shift Leader
ing													
Deflashing		70.01	—	Knife width	---	Max 10 mm Width	Visual inspection	1 No	Initial		Work Instruction for Deflashing - PDN / V1 / 06/ W105	Reaction Plan - QA/V/24/P	Operator
		70.02	Undercut	—	---	Not allowed	Visual inspection	3 No's	Initial			Reaction Plan - QA/V/24/P	Operator
		70.03	Surface Damage	—	---	Not allowed	Visual inspection	3 No's	Initial			Reaction Plan - QA/V/24/P	Operator
ooling													
Post Cooling		80.01	—	Process Parameters	---	As per Process Parameter check - PPC / 02 / V105 W1 for Post Cooling - PDN / W1 / 07	Display monitor	One Time	Once in 2 Hours		Process Parameter Check sheet Boring & Welding (F/PDN/02) Shift start up check sheet - Post Cooling - F/PDN/05 5's Check Sheet - F/PDN/19 MES System X Bar R Chart (for Weight)	Reaction Plan - QA/V/24/P	Shift Leader
		80.02	—	Identification of fixture no. & Date & Shift on tank body	---	Fixture No. & Date & Shift should be marked on tank body	Visual	100%	Each lot				Operator
			—	Bar code Pasting & Scanning	---	Each tank should be identified with barcode. & Scanned in MES As per W1 for Bar Coding - PDN / W1 / 08	Visual & MES	100%	Each lot				Operator
		80.03	Weight	—	SC	Weight Specification matrix - PDN/PD/01/P 6300 ± 200 grams	Electronic Weighing Balance	100%	Each lot				Operator
ion													

8. Standardisation

8.1 Documents update:

Action Plan	Target Dt	Resp.	Status
PCRN for Diesel Poka-yoke implementation	31 July 2019	Sayali More	Closed



8. Standardisation

8.1 Documents update:

Action Plan	Target Dt	Resp.	Status
PFMEA for Diesel Poka-yoke implementation	20 Nov 2019	Suryabhan	Closed

		Model: Scorpio/Marazzo														Cross Functional Team			
		Part Name/ Process														Nikhil/ Suryabhan	Hari Naga		
		Opn No.: 6180														Vikrant Ghule	Vikrant Morade		
		Opn Despt.: Fuel tank fitment																	
assis		MIC Name & No.:																	
Requirements	Potential Failure Mode	Potential Effect(s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
2) Tightening of fuel tank bolts.	1) Bolt under torqued/loose. LSP V1	Mfg & Assy: Squeak & rattle (4) End User: Fuel tank will be loose leading to unsafe vehical operation due to vibration (9).	6		Operator forgot to tight the bolt.	1 Pre tightening of all the Bolts up to 3-4 threads 2 Tightening of the blind bolts, till the tool shuts off. 3 Tightening of the Fit bolts & the side bolts after that till the tool shuts off. 4 Paint marking for tightening confirmation. 5 Buy off Visual check with marking	2	Self certification.	7	84	Poka yoke started for bolt loose or bolt not fitted.	MP/SM Nov'19	Closed	8	2	7	84
					2) Improper power tool calibration	Calibration as per frequency decided in control plan	2	Sticker for due date for calibration on tool	7	100							
	2) One Bolt not fitted.	Mfg & Assy: Rework at post station (5) End User: Vibration will cause fuel tank bolt loose and noise (8) BUY OFF concern Aug'17	6		1) Holes are not matched / aligned.	Awareness & Training to operator	5	At the same stage At chassis Buy-off.	7	105	Poka yoke started for bolt loose or bolt not fitted.	MP/SM Nov'19	Closed	8	2	7	84
Rev Date		Revision Summary							Reason for Revision										
20.11.19		Poka yoke started for bolt loose or bolt not fitted.							PCRM/A000/STC/F/2010/786										

Mahindra & Mahindra Ltd.

Process Potential Failure Mode & Effect Analysis.

Rev. No.: 40 | Doc No.: PFMEA/SPU/SCORPIO/6180

Rev. Date: 20.11.19 | FMEA Date (Orig.): 12/01/2002

8. Standardisation

8.1 Documents update:

Action Plan	Target Dt	Resp.	Status
Control Plan for Diesel Poka-yoke implementation	20 Nov 2019	Suryabhan	Closed

Plant : Kandi	Model :Scorpio	Mahindra & Mahindra Ltd. Control Plan	Cross Functional Team	
Part Refs.: - 3M0845CNC0315EBR	Part Name/ Process: Chassis		Vikrant Ghole	Sopan More
Latest Change Letter:-	Doc No.:	Prototype☐ Prelaunch☐ Production☒ Critical Char : (CC)	Suryabhan Sur	Taj Shaikh
Aggregate No.:	Aggregate Part Description	Models Name : Scorpio Assy. Reviewed by :Vikrant Ghole Rev. No.: 112 Significant Char : (SC)	Pravin Patil	Vikrant Morade
PDF Shop :Scorpio/Xylo TCF	Next Aggregate No.:	Cell Name :Chassis Approved by :Manoj Jadhav Rev. Date :20.11.19	CP Date(Orig) :15.11.2005	

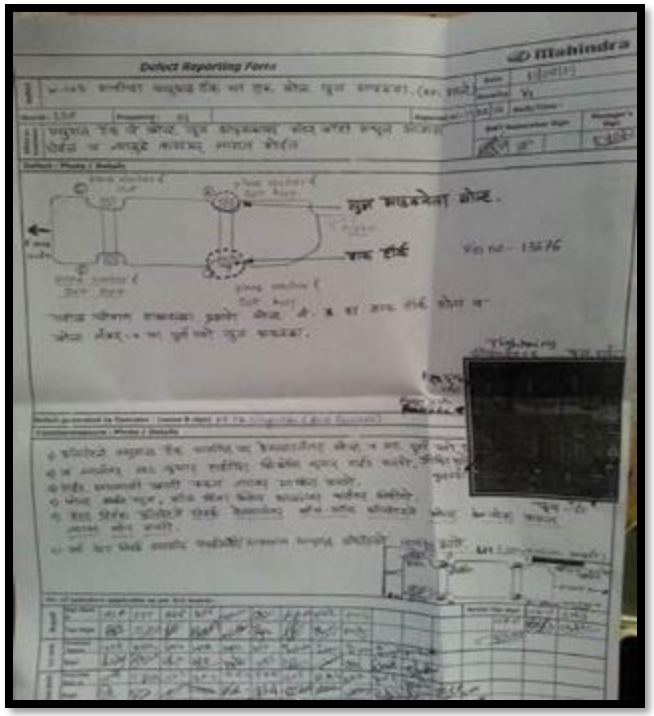
Opn No.	Operation Description	Machine, Device, Jig, Tools Description	No.	Characteristics: Product	Characteristics: Process	Sp. Char. Class	Method: Product/ Process Specification.	Method: Evaluation Measurement Tech.	Method: Sample Size	Method: Sample Freq.	Method: Control Method Record	Method: Control Method Resp.	Reaction Plan
3180	Fuel Tank Assy Fitment On Frame		1	Fuel tank selection as per model	Assy of fuel tank as per Model from the build sheet		Sequenced trolley , check Buildsheet, orange sticker for S3, blue colour for S5-S7	Visual	100%	Daily	Traveller card	Associate on that stage	Replace
			2	No strap hanging			Cutting of strap after placement	Visual	100%	Daily	Traveller card	Associate on that stage	Replace
			3	Stone guard selection as per model	collection of stone guard as per build sheet		Small Size for 2WD Big Size for 4WD / Petrol For S/C DIC-2900N 4 Holes for Xylo Quanto S2 3 Holes for S4-S10	Visual	100%	Daily	Traveller card	Associate on that stage	Rectify
		Torque controlled nut runner	4	Torque	Poka yoke for tool torquing	CC	30 ± 5 Nm	Torque Audit with ACTA unit	3	Daily	SPU/TCF/MQA/F105	Associate - MQA	As per SOS NSK/STCF/MQA/23
			5		Poka yoke for tool torquing Calibration of nut runner	CC	30 ± 5 Nm	Calibration with ACTA unit	4	Once in 2 months	SPU/TCF/R/113	Associate maint.	3)As per SOS NSK/STCF/MQA/23
3190	Front Axle Insulator & Bolt Pre Assy (4 WD Only)		1	Assembly of rubber bush, metal sleeve & washer			Complete insertion	Visual	100%	Daily	Nil	Associate - TCF	Error in Traveller card & Inform to officer.

Rev No.	Rev Date	Revision Summary	
112	20.11.19	Fuel tank bolt poka yoke started.	FCRN

8. Standardisation

8.2 Training:

Action Plan	Target Dt	Resp.	Status
On job Training through DRF given to all the stage operators about the tightening sequence	10 Aug 2019	Mandar	Closed



8. Standardisation

8.3 Additional audit / Check:

Permanent checkpoint added & Reaction plan updated in control plan:

PRODUCT DIMENSIONAL CONTROL PLAN

Prototype | Pre-launch | Production

Key Contact (Tel): Jayant Bhavikatti (9011291309)

DATE (ORIG): 15/01/2015 | DATE (REV): 12/10/2014

Part No/Latest Change Level:PR7

Cross Functional Team

Quality: Jayant,Jayant Zinna, Vivek,Vijay,Anand,Amar; Engineering: Amardeep,Kanishek,Amit,Atul,Vineswara

Production:Sandeep,Yogesh,Balaji,Sunil,Pulung,Suryakant,Inamder,Ravindra,Tushar,Chaitanya

Part Name/Description:REAR MODE ASSY. (T-40 (Model-2WD Dash /Non Dash/UCON/4WD)

customer SQA Approval: 02/08/2014

customer quality approval date(if Reqd)

Supplier/Plant: Cosma International(India) Pvt Ltd | Supplier Code: DUN3-477535834

Other Approval Date(if Reqd)

other approval date(if Reqd)

Part Process No.	Process Name / Operation Description	Machine,Device,Jig,Tools/ Supp.for Mfg	Part Number	Quantity per Vehicle	Characteristics: No.	Characteristics: Product	Characteristics: Process	SC/CC	Methods: Product / Process /Specification	Methods: Evaluation /Measure	Methods: Sample: Size	Methods: Sample: Frequency	Methods: Control Method	Error Proofing	Responsibility	Reaction Plan
Rear Mode Assy (T-40)	Mid Welding	Welding M/C (LWCON) Welding Fixture # FIXT-M-120	1.8206ABC00290N 2.8206ABC00300N 3.8206ABC00260N 4.8206ABC00220N 5.8206ABC00180N 6. Out Put Of rear Mod Zone 1	1 1 1 1 2 1	1	Cosmetic Quality			No Missing Functional Hole and Slots	Visual	All	Visual	Pins & Sensors	NA	Prod. Operator	In case of Fault Detection Quarantine the material & Inform to production supervisor/QA supervisor/maintenance Engineer/Weld Engineer
									Part should free from all welding defects-Weld Off, Surface Crack, Burn Through, Missing Weld, Weld Undercut, Porosity, Crater, Weld Spike,Weld Notch, Other Weld Defect			NA	First Off Last Off Inspection Report & Witness mark for all remaining (Doc. No. QA-IRM&M-K 40)			
					2	Functional Quality Presence of Weld Seam	NA	NA	24 Nos	Visual	All	All	First Off Last Off Inspection Report (Doc. No. QA-IRM&M-K 40)	PLC & Robot Prog.	Prod. Operator	
					3	Weld Seam Length			Ref Weld map Sheet(DOC NO.-CI-F-ELD-009)	Scale/Thread	1	Weekly	Weld Length Report (Doc. No. -CI-F-QA-WSL-074)		QA Inspector	
					4	Weld Leg length			≥ 5 mm	Weld Destructive Testing	1	Weekly/Per Build	Weld tear down Analysis Report (Doc. No.- CI-F-QA-WTA-072)		Q. Auditor/Lab Technician/ Weld Engg	
					5	Throats Thickness			≥ 3.75 mm							
					6	Penetration			≥ 0.25 mm							
					7	Poka Yoke Verification	Ref Error Proofing List-CI-F-MMT-LEP-020		For Child Part & Nut Missing Verification	Poka Yoke Sample /PLC	1	Daily	Daily Start off Check Sheet (Doc No-CI-F-PRDN-OPD-003)	PLC	Prod. Operator	
					8	Masking Of Fuse Face Nut	NA	NA	2 mm	Visual	All	Daily		NA	Prod. Operator	
					9		Welding Condition		Ref Process Parameter Sheet	Pendet Display	1 time	Weekly	Process Parameter Sheet	Password Protected	Maint. Engg. /Weld Engg	
					10		Gas Flow rate	NA	18-24 Ltr/Min	Gas flow Meter	1 time	Daily	First Off Last Off Inspection Report (Doc. No. QA-IRM&M-K 40)	Alarm System	Prod. Operator	
							Air Pressure	NA	5-7 Bar	Pressure Gauge	1 time	Daily				
					11		Tip change Cycle	NA	Cleaning Tip by appointed period	Visual	1 time	Start	Daily start off Check Sheet (Doc No-CI-F-PRDN-OPD-003)	NA		
					12		Jig condition		No loosen bolt and no broken pin., No spatters on nozzles	Visual	—	Start				

9. Standardisation

9.1 Conclusion:

Concern is resolved by incorporating a separate Process of Safe guarding the Fasteners from Foreign particles like Weld Spatter at Sub Assy and Main Frame Assembly.

Automatic Tool selection mechanism helps to eliminate the possibility of wrong torque delivery while fuel tank tightening and Poka-yoke ensures no torque skipping and required torque delivery to each bolt.

9.2 Horizontal Deployment:

Implemented for All the Fastener of the Cosmo Magna Chassis.

9.3 Lessons Learnt:

Things gone wrong: Just Dimensional Accuracy for Part fitment can’t be the only solutions .

Thing gone right: CFT approach helped for concern resolution.

QC story approach is an effective and accurate way to solve complex problems.

9. Summarise & Plan future actions

Future Actions:

Enclosing all the Fasteners of the Chassis Frame at Cosmo Magna by Special Enclosure to avoid the Welding Spatters Accumulation .

Action planned :- Special Common Enclosure for the Main Frame Welding Process.

Responsibility :- Mr. P S Kulkarni SQA

Target Date :- 28th Feb 2020.